

The Move Layer

Orbit Opening, Transport, and the Inverse Problem in BBP Relation Spaces

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Abstract

The Transform Layer established how a known BBP relation moves under subdivision. This paper establishes what a move is permitted to open, proves that the apparent "creation ledger" of the previous phase is not a model but the transport null itself, and isolates the single unresolved structural question that remains.

Three results are derived rather than fitted. First, the transported periodic subspaces satisfy $T_a \cap T^b = T_{\gcd(a,b)}$, making the divisor lattice distributive and inclusion–exclusion exact; consequently the Möbius inversion of a measured dimension sequence carries zero empirical content and only the excess above it is observable. Second, the octal rational kernel is the transported order-3 subgroup of $\mathbb{Z}/q$, giving $K_q = \gcd(q,3)$ exactly on all six measured subdivisions with carrier positions predicted from group structure alone. Third, the orbit channels are installed by the factorization of $b-1$, not by the move; the move only selects among them, and the least common multiple of the orbit-opening moves is the first frame at which two independently opened channels can interact.

Five candidate laws are recorded as falsified, each killed by measurements already present in the corpus rather than by new computation. The paper closes with one preregistered experiment, its transport null, its legal outcome range, and the measurement record required for the result to be interpretable.

1. Status Ledger

Every claim below carries a status. Nothing is promoted on the strength of agreement between readers of the same case set.

Status	Claim
FORCED	Subdivision $n \rightarrow nq$ acts by zero insertion: $W_k = (1/q) \cdot w_{(k \bmod n)}$
FORCED	$T_a \cap T^b = T_{\gcd(a,b)}$; $\langle T_a, T^b \rangle = T_{\text{lcm}(a,b)}$ on the periodic subspaces

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Status	Claim
FORCED	Inclusion–exclusion is exact over the divisor lattice; the Möbius ledger of the transport-generated sequence is the transport null
FORCED*	Minimal Interaction Frame: $q_{\text{int}} = \text{lcm}(q_1, \dots, q_k)$ — forced GIVEN the identified opening moves
CONFIRMED	Octal rational kernel $K_q = \gcd(q, 3)$, $6/6$ on $q = 1, 3, 5, 6, 9, 15$
PULL	Reader aperture is a modulus, not a bound (proposed mechanism for the above)
CONFIRMED	Carrier positions = image of the order-3 subgroup of $\mathbb{Z}/q$, $4/4$
CONFIRMED	Octal $R = 5$ exactly at $q \in \{3, 5\}$, the prime factors of $b-1 = 15$
CONFIRMED	Reader ladder at $(16, 24)$: $\mathbb{Q} \rightarrow 3$, $\mathbb{Q}(2^{\{1/6\}}) \rightarrow 3$, $\mathbb{Q}(2^{\{1/6\}}, \sqrt{3}) \rightarrow 5$
PULL	Orbit opening: $p \mid q$ and $p \mid (b-1)$ selects the p -channel (octal-supported, lineage-specific)
OPEN	R_{15} at $(16, 120)$ — the interaction residual $E_{\text{int}}(15) = R_{15} - 9$
OPEN	Hexagonal R_3, R_7, R_{21} ; repeated-prime behaviour at $3^2 \mid 63$
OPEN	Hexagonal $K_6 = 4$ — incompatible with every tested candidate law, in both directions
OPEN	Channel asymmetry: $\mathbb{Q}$ resolves the 3-channel but not the 5-channel though both are quadratic
DEAD	$q \bmod 6$; conditional $\gcd(q, 6)$; $\gcd(q, 12)$ as a universal hexagonal law
DEAD	ϕ -aperture law $K_D(q) = \sum_{\{d \mid q, d \leq D\}} \phi(d)$
DEAD	Move-invariance: c^K as a function of q alone across lineages
DEAD	Ramification rule: $p \mid n \Rightarrow c_p = p - 1$
DEAD	Parity / alternation rule for the hexagonal sequence
RETRACTED	The earlier rank-10 R_{120} result (reconstruction defect)

2. Central Thesis

Subdivision is not the creation mechanism. It is the transport mechanism.

The seed factorization determines which orbit channels exist. A subdivision move selects which of those channels are active in the destination frame. Once a relation exists on a channel, every later subdivision transports it exactly by zero insertion. The first frame at which two independently opened channels coexist is the first frame capable of testing their interaction.

orbit structure $\rightarrow$ move $\rightarrow$ opening $\rightarrow$ transport $\rightarrow$ interaction

The protagonist is the move, not the dimension. The previous phase treated $\dim R$ as the object of study and produced four successive formulas in q , all of which died. The reason they died is structural rather than accidental: a dimension at q is an accumulation over the divisor lattice below q , and no arithmetic function of q can express an accumulation whose overlap terms depend on $\gcd$ s.

3. The Move as Primitive

Write M_q for the subdivision $n \rightarrow nq$. Three objects must be kept distinct: the source frame, the move, and the destination frame. The transform theorem is a statement about the move.

$$M_q : (b, n) \rightarrow (b, nq), \quad W_k = (1/q) \cdot w_{(k \bmod n)}$$

The transport operator is indexed by q . The content transported is not. This distinction was established the hard way: the same subdivision factor $q = 6$ realises dimension 4 in the hexagonal lineage and dimension 3 in the octal lineage. Any claim that a measured quantity is "a function of the move" must therefore specify which lineage the move acts on.

Corollary (identity move). $q = 1$ is the identity $n \rightarrow n$ and creates nothing by definition. The entry c_1 in any Möbius-inverted ledger is therefore a state, not a move, and must not be counted alongside the move entries. Cross-lineage agreement at $q = 1$ is category-trivial.

4. Transport Algebra

Transport by zero insertion sends a relation at frame nd to a relation at frame nq that is nd -periodic. The image T_d is therefore the space of nd -periodic vectors inside the relation space at nq .

Theorem 4.1. For $d, e \mid q$, $T_d \cap T_e = T_{\gcd(d,e)}$ and $\langle T_d, T_e \rangle = T_{\text{lcm}(d,e)}$.

Proof sketch. A vector on $\mathbb{Z}/nq\mathbb{Z}$ invariant under shift by nd and by ne is invariant under the subgroup they generate, which is $\gcd(nd, ne)\mathbb{Z}/nq\mathbb{Z}$. The inverse transform pulls an nd -periodic relation at nq back to a relation at nd . The lattice of transported subspaces is therefore isomorphic to the subgroup lattice of a cyclic group.

Verification (octal seed $n = 8$):

```
q=15 T_3 n T_5 : dim=8 8*gcd=8 True
q=6 T_2 n T_3 : dim=8 8*gcd=8 True
q=12 T_4 n T_6 : dim=16 8*gcd=16 True
q=9 T_3 n T_9 : dim=24 8*gcd=24 True
q=24 T_8 n T_12 : dim=32 8*gcd=32 True
q=30 T_6 n T_10 : dim=16 8*gcd=16 True
```

Corollary 4.2. Divisor lattices are distributive, so inclusion–exclusion over them is exact:

$$\dim(\sum_{d \mid q} T_d) = \sum_{d \mid q} c_d, \quad c = \mu * (\dim T)$$

```

q=6  dim(sum T_d)=48  sum c_d=48  True
q=9  dim(sum T_d)=72  sum c_d=72  True
q=12 dim(sum T_d)=96  sum c_d=96  True
q=15 dim(sum T_d)=120 sum c_d=120  True
q=24 dim(sum T_d)=192 sum c_d=192  True
q=30 dim(sum T_d)=240 sum c_d=240  True

```

5. The Creation Ledger Is the Transport Null

This is the most consequential retraction of the phase. The previous work Möbius-inverted a measured dimension sequence, obtained $c = (1,1,2,2,0,0,0,0)$, and read the four nonzero entries as creation events and the four zeros as confirming predictions.

Corollary 4.2 shows this reading is empty. The divisor-sum identity is forced by the transport algebra, so every prediction the ledger makes about the transported part is a theorem, not a claim. The four zeros were not four passed tests. The ledger never made a testable statement at all.

Stated exactly: the Möbius ledger of the transport-generated sequence is the transport null. This is not a licence to Möbius-invert an arbitrary measured sequence and call the result a null — that holds only when the sequence is interpreted under the same transport structure and the same reader (Rule 11.2).

What survives is the definition of the observable.

$$F(q) = \dim \sum_{\{d|q, d < q\}} T_d(X_d) \quad E(q) = X_q - F(q)$$

F is the transport null: everything the ancestry already guarantees. E is the excess: the only quantity that can carry information about a frame. A measurement returning $X_q = F(q)$ confirms transport and nothing further. This applies to K and to R separately, with F computed in the same reader as X.

Practical consequence: measuring a frame whose ledger value equals its transport floor is not an experiment. Under this criterion, $q = 5, 7, 8, 9, 10, 15$ and 16 in the hexagonal band are all information-free with respect to the ledger, and were incorrectly queued as discriminators during the previous phase.

6. Orbit Structure Precedes Dimension

The channels available to a lineage are installed by the seed, not by the move. For the octal lineage the factorization of $x^8 - 16$ partitions the residue classes mod 8:

Classes	Factor	Channel
$\{0,1,4,7\}$	unit support	—
$\{2,6\}$	$x^2 + 2$	3-orbit
$\{3,5\}$	$x^2 + 2x + 2$	5-orbit

The primes carried by these channels are exactly the prime factors of $b - 1$:

$$b-1 = 15 = \{3, 1, 5, 1\}$$

$q=1$ $R=1$ q a prime factor of $b-1$? False rule->1 OK

$q=2$ $R=1$ q a prime factor of $b-1$? False rule->1 OK

$q=3$ $R=5$ q a prime factor of $b-1$? True rule->5 OK

$q=4$ $R=1$ q a prime factor of $b-1$? False rule->1 OK

$q=5$ $R=5$ q a prime factor of $b-1$? True rule->5 OK

$q=7$ $R=1$ q a prime factor of $b-1$? False rule->1 OK

This explains the long-standing $q = 7$ negative result directly: $7 \nmid 15$, so there is no 7-channel in the octal seed for the move to select. The move cannot invent a channel; it can only reach one that the seed installed.

Status: PULL, lineage-specific. The rule is supported on six octal frames and is not asserted for the hexagonal lineage, where $b - 1 = 63 = 3^2 \cdot 7$ has a different structure.

7. The Rational Kernel Is a Subgroup

The strongest confirmed result of the phase. The octal rational kernel was measured at six subdivisions, and the recorded carrier positions identify it exactly.

q	K_q measured	$\gcd(q, 3)$	carriers recorded	multiples of $q/3$
1	1	1	—	—
3	3	3	$\{0, 1, 2\}$	$\{0, 1, 2\}$
5	1	1	—	—
6	3	3	$\{0, 2, 4\}$	$\{0, 2, 4\}$
9	3	3	$\{0, 3, 6\}$	$\{0, 3, 6\}$
15	3	3	$\{0, 5, 10\}$	$\{0, 5, 10\}$

The carriers are the image of the order-3 subgroup of $\mathbb{Z}/q$ under the subdivision embedding. The heights (8, 32, 48) are invariant across all four frames; only the spacing changes. The rational reader is therefore not resolving "three relations" — it resolves one subgroup, and reports its cardinality.

$$K_q = \sum_{d \mid \gcd(q, 3)} \phi(d) = \gcd(q, 3)$$

Observation 7.1 (measured). The octal rational reader behaves as an order-3 modulus: it resolves the order- d sector when $d \mid 3$ and returns nothing otherwise, on all six measured subdivisions, with carrier support matching in every case. [CONFIRMED]

Conjecture 7.2 (reader modulus). The aperture of a reader is a modulus rather than a bound: a reader resolves the order- d sector precisely when $d \mid m$ for some modulus m determined by its field. [PULL — proposed mechanism for Observation 7.1, not derived]

The conjecture is not supplied with a derivation here, and the reason is worth recording as an open problem rather than a gap. The two octal channels correspond to distinct quadratic fields:

$$\{2,6\} \quad x^2 + 2 \quad \text{roots } \pm i\sqrt{2} \rightarrow \mathbb{Q}(\sqrt{-2}) \quad \text{degree } 2$$

$$\{3,5\} \quad x^2 + 2x + 2 \quad \text{roots } -1 \pm i \rightarrow \mathbb{Q}(i) \quad \text{degree } 2$$

Open problem 7.3. Both channel fields are quadratic over $\mathbb{Q}$, so no degree argument separates them — yet the rational reader resolves the 3-channel and is blind to the 5-channel. Until that asymmetry is explained, $m = 3$ is a measured value and not a derived one.

This reframes an earlier observation. The rule "twisted channels exist if and only if $3 \mid q$ " is not a placement law about twisting; it is the statement that $\mathbb{Q}$ can see only the order-3 sector, so it sees something exactly when $\mathbb{Z}/q$ has one.

Consequence: the octal rational reader is closed. $K_q = \gcd(q, 3)$ for every measured q , and the reader saw everything it will ever see at $q = 3$. No further K -measurement in this lineage can produce new information. Every remaining question lives in R .

8. R Is Not a Subgroup Count

$R_3 = R_5 = 5$ while $\gcd(3, 5) = 1$. No modulus-based law can return equal values at coprime indices, so R is not counting subgroup structure of $\mathbb{Z}/q$. K and R are different species of object, and the inclusion $K \subset R$ is not an inclusion of like things: R is a relation space over a wider field, and only its intersection with $\mathbb{Q}^N$ comes out subgroup-shaped.

The reader ladder recorded at (16, 24) shows the separation directly:

$$\begin{array}{ll} \mathbb{Q} & \rightarrow 3 \\ \mathbb{Q}(2^{1/6}) & \rightarrow 3 \quad \text{radical alone adds nothing} \\ \mathbb{Q}(2^{1/6}, \sqrt{3}) & \rightarrow 5 \quad \text{the two extra directions require } \sqrt{3} \\ \overline{\mathbb{Q}} & \rightarrow 5 \end{array}$$

The two additional directions are the antisymmetric / argument component and enter through the cyclotomic side. This ladder is what rescued $R_3 = 5$ from being read as a saturation artifact — without it, the number 5 appearing at two structurally different moves would have been indistinguishable from an instrument ceiling.

9. The Minimal Interaction Frame

Proposition 9.1 [FORCED given the opening-move model]. Let a lineage have independently opened orbit families with minimal opening moves $q_1, \dots, q_k$. The smallest subdivision index at which all of them coexist is $q_{\text{int}} = \text{lcm}(q_1, \dots, q_k)$. This identifies the first frame at which interaction among the

corresponding transported subspaces can be tested. It does not by itself imply independence, additivity, or the absence of further structure. The lcm computation is forced once the opening moves are identified; identifying 3 and 5 as the octal opening moves is a separate, lineage-specific empirical claim (§6, PULL).

octal $b-1=15=\{3:1, 5:1\}$ opening moves [3, 5] $q_{int}=15$ $N=120$
 hexagonal $b-1=63=\{3:2, 7:1\}$ opening moves [3, 7] $q_{int}=21$ $N=126$

$N = 120$ was never merely a large frame. It is the minimal frame at which the two independently opened octal channels can meet. That is why it is the only octal frame whose measurement carries information about their interaction, and — by an unfortunate coincidence — it is also the frame where the reconstruction defect occurred.

Replication warning. The number of genuinely independent interaction tests a lineage offers is governed by $\omega(b-1)$, the count of distinct prime factors. Octal offers one minimal two-orbit frame. Later multiples of 15 are descendants of that first interaction, not independent repeats. If the result at $N = 120$ is ambiguous, the octal lineage cannot supply a second minimal interaction frame from the same two orbits.

The hexagonal lineage is therefore not a convenience but a structural requirement. It is also not a replicate: $3^2 \mid 63$ means the hexagonal 3-channel has a repeated-prime structure absent from octal, and hexagonal $q = 9$ may be a second-stage opening within the 3-primary structure rather than transport of $q = 3$. Marked OPEN — this is the leading repeated-prime hypothesis, not a result.

10. The Inverse Problem

The forward reading and the backward reading must close on each other. This is the paper's principal methodological proposal and the reason the transform is worth having as a theorem rather than a procedure.

Direction	Chain
Forward	$(b, n), q \rightarrow \text{channels of } b-1 \rightarrow T_q(\text{orbit}) \rightarrow R_q$
Backward	support at frame $\rightarrow$ occupied orbit classes $\rightarrow$ primes $\rightarrow$ opening moves $\rightarrow b-1$

Proposition 10.1 (orbit reconstruction from support). If a measured relation's support decomposes into identifiable orbit classes of the seed factorization, the occupied classes determine candidate prime channels; the minimal divisor levels at which those channels occur define candidate opening moves; their least common multiple is the minimal interaction index compatible with the observed channels.

Limitation, stated with the proposition: this recovers candidate structure from support. It does not establish that the recovered channels are independent, that no unobserved channels exist, or that the destination dimension is the sum of their dimensions.

For the octal lineage the loop is concrete. Forward: $b - 1 = 15$ gives channels 3 and 5, opening moves 3 and 5, interaction index $\text{lcm}(3,5) = 15$. Backward: support at $N = 120$ landing in classes $\{2,6\}$ and $\{3,5\} \bmod 8$ recovers

$x^2 + 2$ and $x^2 + 2x + 2$, hence primes 3 and 5, hence opening moves 3 and 5, hence lcm = 15. The loop closes on 15 from both ends.

This is why the R_{120} measurement must record support and orbit decomposition, not only a dimension. A single integer can agree with a prediction by accident; a dimension together with a support decomposition that independently reconstructs the seed factorization cannot.

10.2 Forward / Inverse Closure Table

Every row is a claim that the same object is reachable from both ends. A row is closed only when the inverse reading independently reconstructs the forward premise.

Object	Forward reading	Inverse reading	Status
seed	$b-1 \rightarrow$ channel set	support $\rightarrow$ factorization of $b-1$	CONFIRMED
$q = 3$	$3 \mid (b-1) \Rightarrow$ 3-channel opens	$\{2,6\} \rightarrow x^2+2 \rightarrow p = 3$	CONFIRMED
$q = 5$	$5 \mid (b-1) \Rightarrow$ 5-channel opens	$\{3,5\} \rightarrow x^2+2x+2 \rightarrow p = 5$	CONFIRMED
$q = 7$	$7 \nmid 15 \Rightarrow$ no channel	no orbit class occupied	CONFIRMED
$q = 6, 9$	transport of the 3-channel	support $\rightarrow$ 3 only	CONTROL
$q = 15$	$\text{lcm}(3,5) = 15 \Rightarrow$ both present	support $\rightarrow \{3,5\} \rightarrow \text{lcm} = 15$	OPEN
hex $q = 21$	$\text{lcm}(3,7) = 21$	support $\rightarrow \{3,7\} \rightarrow \text{lcm} = 21$	FUTURE

The loop closes on 15 from both ends for the octal lineage — forward from the factorization of $b-1$, backward from the occupied residue classes. Rows marked OPEN and FUTURE are exactly the two interaction experiments of §13.

11. Reader Protocol and Headroom

Every measurement in this program is a readout. There is no measurement of an object. R is not "the object" — it is the widest reader that has been run.

Rule 11.1 (headroom). Repeated identical output under structurally different inputs is evidence of saturation before it is evidence of symmetry.

Record with every measurement:

$$h = M_{\text{instrument}} - O_{\text{observed}}$$

$h = 0$ is a saturation warning and must be labelled at the moment of recording; it cannot be reconstructed later. $h > 0$ removes the hard-ceiling explanation but does not by itself establish that the invariant is structural. This rule was derived after it had already fired twice undetected — once on the apparent K-ceiling of 4, once on $R_3 = R_5 = 5$.

Rule 11.2 (reader consistency). Möbius inversion of a sequence measured under different readers is arithmetic on incommensurable quantities. It will return a clean nonnegative integer vector and mean nothing.

The free diagnostic: under a constant reader, rational transport forces $d \mid q \Rightarrow K_d \leq K_q$. A violation proves reader inconsistency at no computational cost. Both measured sequences pass:

hex: divisor-monotonicity violations = NONE

oct: divisor-monotonicity violations = NONE

Rule 11.3 (oversample, then restrict exactly). Do not run two independently bounded searches in two fields. Run one wide solve and take the exact restriction:

$$K_Q = V_F \cap \mathbb{Q}^N$$

Not a Euclidean projection, which can manufacture vectors that are not valid integer relations. The (16,12) result establishes why this matters: increasing the coefficient bound by 10^4 changed nothing, while adjoining one field element exposed the relation. Capacity and dimension are separate axes, and a wide solve with exact restriction obtains both readings in one coordinate system.

Rule 11.4 (case-set diversity). Concordance between multiple readers of the same digest is not independent validation. Two routes are independent only if the case set spans the hypothesis under test. Applied to a multi-model review network, agreement among models given the same summary is saturation of the shared bottleneck, not confirmation.

12. Falsification Log

Each of the following was killed by data already present in the corpus. No new computation was required for any of them; in each case the falsifying measurement predated the hypothesis.

Killed claim	Falsifier
ϕ -aperture: $K_D(q) = \sum_{\{d \mid q, d \leq D\}} \phi(d)$	Range at $q = 6$ over all D is $\{1, 2, 4, 6\}$; measured $K_6 = 3$
Move-invariance: $c^K = f(q)$ across lineages	$K_6 = 4$ hexagonal vs $K_6 = 3$ octal
Ramification: $p \mid n \Rightarrow c_p = p - 1$	Octal $q = 3$ has $3 \nmid 8$ yet $c_3 = 2$
Reader deficit $\Delta^K(p) = \phi(p)$	Predicts $c_2 = 3$ against a measured 1
Parity / alternation: $K_9 = K_{15} = 1$	Transport forces $K_q \geq K_3 = 3$ whenever $3 \mid q$
$\gcd(q, 12)$ as a universal hexagonal law	Any modulus law forces $K_q \mid q$; hexagonal $K_6 = 4$ and $4 \nmid 6$
"Native means outside the inherited support"	Superseded by support excess (Transform Layer)
"Two coefficient bounds are two routes"	Bound and field are separate axes; (16,12)

One measurement resists every candidate. Hexagonal $K_6 = 4$. Any modulus law forces $K_q \mid q$, and $4 \nmid 6$. $\gcd(q,12)$ predicts 6; the octal-style subgroup law predicts 3; the ϕ -aperture predicts 4 but is dead on other grounds. The frame $(6_4,3_6)$ is unaccommodated by every tested candidate, in both directions simultaneously and is flagged as the corpus's single anomalous datum. Its most likely resolution is that $q = 6$ in the hexagonal lineage is simply not the same move as $q = 6$ in octal, because $6_3 = 3^2 \cdot 7$ installs a different channel structure than $1_5 = 3 \cdot 5$.

13. The Preregistered Experiment

One experiment, two integrity controls, declared before the run.

13.1 Controls (not discriminators)

R_6 at $(16,4_8)$ and R_9 at $(16,7_2)$. Both frames are already instantiated. For $q = 6$ the divisor sources are $\{1,2,3\}$ with $T_2 \cap T_3 = T_1$, giving $F_R(6) = R_2 + R_3 - R_1 = 1 + 5 - 1 = 5$. For $q = 9$ the only nontrivial source is 3, giving $F_R(9) = R_3 = 5$. Every surviving model predicts 5 at both, so neither frame can discriminate between hypotheses. They are reader-integrity checks and must be labelled as such so that a returned 5 is not later cited as evidence for a model it cannot support.

Legal range: $R_6 \geq 5$ and $R_9 \geq 5$. A value below 5 is impossible under transport and indicates a broken reader or reconstruction, not a finding.

13.2 The experiment

R_{15} at $(16,12_0)$, the minimal two-orbit interaction frame. Transport null:

$$F_R(15) = R_3 + R_5 - R_1 = 5 + 5 - 1 = 9$$

Observable:

$$E_{\text{int}}(15) = R_{15} - 9$$

Outcome	Reading
$E_{\text{int}} = 0$	The two orbit families are independent apart from the seed; transport plus inclusion–exclusion accounts for the frame completely
$E_{\text{int}} < 0$	The families collide beyond their inherited intersection; gcd-based inclusion–exclusion fails for the realized R even though it is exact for the periodic subspaces
$E_{\text{int}} > 0$	Structure exists at the interaction frame that neither opened orbit supplies — the first measured instance of something made rather than moved

13.3 Required measurement record

A dimension alone is not interpretable. Record, for every R measurement:

reader field (exact construction, not "a wider reader")

M ambient / instrument maximum

O returned rank

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

$h = M - O$ headroom; $h = 0$ is a saturation flag
 symmetric / antisym the 3+2 signature decomposition
 transported basis vectors identified AS transported, not inferred
 new directions residual after removing the transported basis
 support occupied residue classes mod n
 orbit decomposition which channels of $b-1$ the support recovers

The transported vectors must be identified explicitly in the output rather than inferred from a dimension count. Without that, excess and a re-found copy of the inherited family are indistinguishable in the log — which is precisely why (16,48) and (16,72) remained ambiguous for the whole of the previous phase despite both frames already existing.

13.4 Dependency order

$$R_6, R_9 \text{ (controls)} \rightarrow R_{15} \text{ (experiment)} \rightarrow R_3^{\text{hex}}, R_7^{\text{hex}} \rightarrow R_{21}^{\text{hex}}$$

The hexagonal arm is not optional. Octal supplies exactly one minimal interaction frame; without the second lineage, $E_{\text{int}(15)}$ is an unreplicable result carrying the weight of the entire structural claim. The hexagonal interaction null is $R_3^{\text{hex}} + R_7^{\text{hex}} - 1$, and none of its inputs has been measured. Those two frames — $N = 18$ and $N = 42$ — are smaller than anything in the octal R table.

14. What This Paper Establishes

Established:

- Transport algebra: intersections at gcd, joins at lcm, inclusion–exclusion exact.
- The Möbius creation ledger is the transport null; only excess is observable.
- The octal rational kernel is the transported order-3 subgroup: $K_q = \gcd(q, 3)$, $6/6$, with carrier positions predicted from group structure.
- Reader aperture is a modulus, not a bound.
- Orbit channels are installed by the factorization of $b - 1$; the move selects among them.
- Minimal interaction frame at $q_{\text{int}} = \text{lcm}$ of the opening moves.
- The headroom, reader-consistency, exact-restriction, and case-set rules.

Not established:

- Whether the two octal orbit families remain independent when they first meet.
- Whether the hexagonal repeated prime 3^2 produces a second-stage opening at $q = 9$.
- Why hexagonal $K_6 = 4$.
- Any general law for R. R is not a subgroup count and no mechanism for it is proposed here.

The durable output of this phase is not a formula. It is the separation of questions that were previously entangled: what is forced by transport, what a reader can resolve, what a seed installs, what a move selects,

and what — if anything — appears only when two independently opened structures are first placed in the same frame.

Appendix A. Verification Output

All numerical claims in this paper are quoted from the following run. No figure appears in the text that is not present here.

```
[1] T_a n T_b = T_gcd(a,b) (n=8, octal)
q=15 T_3 n T_5 : dim=8 8*gcd=8 True
q=6 T_2 n T_3 : dim=8 8*gcd=8 True
q=12 T_4 n T_6 : dim=16 8*gcd=16 True
q=9 T_3 n T_9 : dim=24 8*gcd=24 True
q=24 T_8 n T_12 : dim=32 8*gcd=32 True
q=30 T_6 n T_10 : dim=16 8*gcd=16 True
ALL: True
```

```
[2] inclusion-exclusion exact over the divisor lattice
q=6 dim(sum T_d)=48 sum c_d=48 True
q=9 dim(sum T_d)=72 sum c_d=72 True
q=12 dim(sum T_d)=96 sum c_d=96 True
q=15 dim(sum T_d)=120 sum c_d=120 True
q=24 dim(sum T_d)=192 sum c_d=192 True
q=30 dim(sum T_d)=240 sum c_d=240 True
ALL: True
```

```
[3] octal rational kernel: K_q = gcd(q,3)
q=1 K=1 gcd(q,3)=1 OK
q=3 K=3 gcd(q,3)=3 OK
q=5 K=1 gcd(q,3)=1 OK
q=6 K=3 gcd(q,3)=3 OK
q=9 K=3 gcd(q,3)=3 OK
q=15 K=3 gcd(q,3)=3 OK
6/6 exact
```

carrier positions = image of the order-3 subgroup of $\mathbb{Z}/q$:

```
q=3 recorded [0, 1, 2] predicted [0, 1, 2] True
q=6 recorded [0, 2, 4] predicted [0, 2, 4] True
q=9 recorded [0, 3, 6] predicted [0, 3, 6] True
```

q=15 recorded [0, 5, 10] predicted [0, 5, 10] True

[4] octal R vs the factorization of b-1

b-1 = 15 = {3: 1, 5: 1}

q=1 R=1 factor? False rule->1 OK

q=2 R=1 factor? False rule->1 OK

q=3 R=5 factor? True rule->5 OK

q=4 R=1 factor? False rule->1 OK

q=5 R=5 factor? True rule->5 OK

q=7 R=1 factor? False rule->1 OK

[5] minimal interaction frames

octal b-1=15={3:1, 5:1} opening [3, 5] q_int=15 N=120

hexagonal b-1=63={3:2, 7:1} opening [3, 7] q_int=21 N=126

[6] transport null at q=15 (octal)

$F_R(15) = R_3 + R_5 - R_1 = 5 + 5 - 1 = 9$

$E_{int}(15) = R_{15} - 9$ [OPEN]

[8] divisor-monotonicity (free reader-consistency check)

hex: violations = NONE

oct: violations = NONE

Appendix B. Correction Log for This Phase

Corrections made during the work that produced this paper, recorded because the pattern of error is itself a result.

Error	Correction
Ledger treated as a model competing with rival formulas	It is the transport null; it never made a claim
"The four zeros are the ledger's empirical content"	Zeros are floor values; the content was one point, then zero
q = 9 and q = 10 queued as discriminators	Both sit at the transport floor; information-free
Transport floor written as max over divisors	Superseded by exactness: the floor is the divisor sum

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com

Error	Correction
Parity rule scheduled for an experiment	Impossible on paper; violates the transport floor at $q = 9, 15$
c^k asserted to be a function of q alone	Killed by K_6 hexagonal vs octal
ϕ -aperture law proposed as exact on 11 points	Killed one turn later by octal $K_6 = 3$
R treated as the object beneath the readers	R is the widest reader run; the stack has no top
" $R_6 < 5$ is impossible" asserted without R_2	R_2 was in the corpus; the assertion was right for the wrong reason
Hexagonal and octal $q = 9$ treated as the same question	$3^2 \mid 63$ but $3^2 \nmid 15$; different moves

Two mechanisms were proposed and falsified within the same working session, each against pre-existing data. That rate is the intended one: a mechanism that can be killed by the corpus should be killed by the corpus, not by an experiment.